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Encapsulation system for a thermal bridge breaker-to-metal liner

Abstract

An appliance includes an outer wrapper, an inner liner, a trim breaker having a wrapper channel that receives a wrapper edge of the outer wrapper and a liner channel that receives a liner edge of the inner liner, and an insulation material disposed within an insulating cavity defined therebetween. A multi-component thermal encapsulation material defines pre-mix, application and sealing states. The pre-mix state is defined by the distinct components of the thermal encapsulation material being separated from one another, the application state defined by the distinct components combined together into an uncured state of the thermal encapsulation material, and the sealing state defined by the thermal encapsulation material disposed within the wrapper and liner channels and surrounding the wrapper and liner edges, respectively, in the sealing state that defines a hermetic seal between the trim breaker and the outer wrapper and the inner liner.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) The present application is a continuation of U.S. patent application Ser. No. 17/835,018 filed Jun. 8, 2022, now U.S. Pat. No. 12,055,344, which is a divisional of U.S. patent application Ser. No. 16/309,734 filed Dec. 13, 2018, now U.S. Pat. No. 11,402,149, which is a national stage entry of PCT/US2016/055161 filed Oct. 3, 2016, all of which are entitled ENCAPSULATION SYSTEM FOR A THERMAL BRIDGE BREAKER-TO-METAL LINER, the entire disclosures of which are hereby incorporated herein by reference.

FIELD OF THE DEVICE

(1) The device is in the field of structural cabinets for appliances, and more specifically, an encapsulation system for attaching a metallic liner and metallic wrapper to a plastic thermal trim breaker.

SUMMARY

(2) In at least one aspect, an appliance includes an outer wrapper, an inner liner, a trim breaker having a wrapper channel that receives a wrapper edge of the outer wrapper and a liner channel that receives a liner edge of the inner liner. An insulation material disposed within an insulating cavity is defined between the outer wrapper, the inner liner and the trim breaker. A multi-component thermal encapsulation material defines a pre-mix state, an application state and a sealing state. The pre-mix state is defined by distinct components of the thermal encapsulation material being separated from one another, the application state defined by the distinct components combined together into an uncured state of the thermal encapsulation material, and the sealing state defined by the thermal encapsulation material disposed within the wrapper and liner channels and surrounding the wrapper and liner edges, respectively, in the sealing state that defines a hermetic seal between the trim breaker and the outer wrapper and the inner liner.

(3) In at least another aspect, a method for forming an insulating cabinet for an appliance includes disposing distinct components of a multi-part thermal encapsulation material into respective dispensing chambers to define a pre-mix state of the thermal encapsulation material. The distinct components of the thermal encapsulation material are delivered from the respective dispensing chambers to a mixing chamber. The distinct components are mixed in the mixing chamber to define an application state of the thermal encapsulation material. The thermal encapsulation material is delivered in the application state to a liner channel and a wrapper channel of a trim breaker. A wrapper edge of an outer wrapper is disposed into the wrapper channel so that the thermal encapsulation material surrounds both sides of the wrapper edge within the wrapper channel. A liner edge of an inner liner is disposed into the liner channel so that the thermal encapsulation material surrounds both sides of the liner edge within the liner channel. The thermal encapsulation material is cured within the wrapper and liner channels to define a sealing state of the thermal encapsulation material, wherein the thermal encapsulation material in the sealing state defines a hermetic seal between the inner liner and the trim breaker and between the outer wrapper and the trim breaker.

(4) In at least another aspect, a method for forming an insulating cabinet for an appliance includes disposing a first component and second components of a thermal encapsulation material into respective first and second dispensing chambers to define a pre-mix state of the thermal encapsulation material. The first and second components of the thermal encapsulation material are delivered from the respective first and second dispensing chambers to a mixing chamber. The thermal encapsulation material is activated by combining the first and second components within the mixing chamber to generate a chemical reaction that defines an application state of the thermal encapsulation material. The thermal encapsulation material is disposed in the application state into a liner channel and a wrapper channel of a trim breaker. A portion of an inner liner is positioned within the thermal encapsulation material in the liner channel. A portion of an outer wrapper is positioned within the thermal encapsulation material in the wrapper channel. The thermal encapsulation material is cured around the portions of the inner liner and outer wrapper within the liner and wrapper channels, respectively, wherein the cured thermal encapsulation material defines a sealing state of the thermal encapsulation material that is characterized by a hermetic seal between the inner liner and the trim breaker and between the outer wrapper and the trim breaker.

(5) These and other features, advantages, and objects of the present device will be further understood and appreciated by those skilled in the art upon studying the following specification, claims, and appended drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) In the drawings:

- (2) FIG. 1 is a front perspective view of an appliance incorporating a thermal trim breaker that is attached to an outer wrapper and inner liner using an aspect of the encapsulation system;
- (3) FIG. 2 is a schematic perspective view of a thermal encapsulation material delivery system for disposing the thermal encapsulation material within a thermal trim breaker;
- (4) FIG. 3 is a schematic cross-sectional view of the trim breaker of FIG. 2, taken along line III-III, and showing deposition of the thermal encapsulation material within the wrapper and liner channels of the thermal trim breaker;
- (5) FIG. 4 is a cross-sectional view of the thermal trim breaker of FIG. 3 showing the thermal encapsulation material disposed within the wrapper and liner channels;
- (6) FIG. 5 is a schematic cross-sectional view showing an outer wrapper and inner liner being placed within the wrapper and liner channels of a thermal trim breaker using the thermal encapsulation material;
- (7) FIG. 6 is a schematic perspective view of a structural cabinet assembled using an aspect of the thermal encapsulation material;
- (8) FIG. 7 is a schematic cross-sectional view of the cabinet of FIG. 6 taken along line VII-VII, and showing the outer wrapper and inner liner disposed within the wrapper and liner channels of the thermal trim breaker;
- (9) FIG. 8 is a perspective view of a wrapper edge of an outer wrapper incorporating a plurality of wrapper protrusions;
- (10) FIG. 9 is an elevational view of the wrapper of FIG. 8;
- (11) FIG. 10 is a schematic cross-sectional view of a structural cabinet assembled using an aspect of the thermal encapsulation material and a plurality of positioning protrusions disposed on the outer wrapper and inner liner;
- (12) FIG. 11 is a schematic flow diagram illustrating a method for forming an insulating cabinet for an appliance using a thermal encapsulation material; and
- (13) FIG. 12 is a schematic flow diagram illustrating a method for forming an insulated cabinet for an appliance using an aspect of a thermal encapsulation material.

DETAILED DESCRIPTION OF EMBODIMENTS

- (14) For purposes of description herein the terms “upper,” “lower,” “right,” “left,” “rear,” “front,” “vertical,” “horizontal,” and derivatives thereof shall relate to the device as oriented in FIG. 1. However, it is to be understood that the device may assume various alternative orientations and step sequences, except where expressly specified to the contrary. It is also to be understood that the specific devices and processes illustrated in the attached drawings, and described in the following specification are simply exemplary embodiments of the inventive concepts defined in the appended claims. Hence, specific dimensions and other physical characteristics relating to the embodiments disclosed herein are not to be considered as limiting, unless the claims expressly state otherwise.
- (15) As illustrated in FIGS. 1-7, reference numeral 10 generally refers to a structural cabinet for an appliance 12, where the structural cabinet 10 can define a vacuum insulated structure or can house various vacuum insulated structures that add thermal functionality to the performance of the appliance 12. According to the various embodiments, the appliance 12 includes an outer wrapper 14, an inner liner 16 and a trim breaker 18, such as a thermal trim breaker 18, that can be attached to define the structural cabinet 10 for the appliance 12. The trim breaker 18 includes a wrapper channel 20 that receives a wrapper edge 22 of the outer wrapper 14. The trim breaker 18 also includes a liner channel 24 that receives a liner edge 26 of the inner liner 16. An insulation material 28 can be disposed within an insulating cavity 29 defined within the structural cabinet 10

inbetween the outer wrapper **14**, inner liner **16** and trim breaker **18**. The trim breaker **18** can be attached to the inner liner **16** and outer wrapper **14** using a multi-component thermal encapsulation material **30** that defines a pre-mix state **32**, an application state **34**, and a sealing state **36**. The pre-mix state **32** is defined by two or more distinct components **38** of the thermal encapsulation material **30** being separate from one another within separate component dispensing chambers **40** of a material delivery mechanism **42**. The application state **34** of the thermal encapsulation material **30** is defined by the distinct components **38** being combined together by an impeller **44** or other mixing device within a mixing chamber **46** of the material delivery mechanism **42**. As the distinct components **38** are mixed together in the mixing chamber **46**, the thermal encapsulation material **30** defines the uncured application state **34** of the thermal encapsulation material **30**. The thermal encapsulation material **30** in the application state **34** is sufficiently fluid to allow for deposition of the thermal encapsulation material **30**, while in the application state **34**, to be disposed within the liner and wrapper channels **24**, **20** of the thermal trim breaker **18**. The sealing state **36** of the thermal encapsulation material **30** is defined by the thermal encapsulation material **30** being disposed, in a cured state, within the liner and wrapper channels **24**, **20** of the thermal trim breaker **18**. In the sealing state **36**, the thermal encapsulation material **30** surrounds the wrapper and liner edges **22**, **26**, respectively, and is cured to define a hermetic seal **48** between the trim breaker **18** and the outer wrapper **14** in between the trim breaker **18** and the inner liner **16**.

(16) Referring again to FIGS. **1-7**, it is contemplated that the thermal encapsulation material **30** in the sealing state **36** can be used to generate the hermetic seal **48** at the trim breaker **18** to allow for an at least partial vacuum **60** to be generated within the insulating cavity **29** of the structural cabinet **10**. The at least partial vacuum **60** can be generated through the expression, expulsion, or other removal of gas **62** from the insulating cavity **29**. This removal of gas **62** creates a pressure differential **64** between the atmosphere around the exterior **66** of the structural cabinet **10** and the at least partial vacuum **60** within the insulating cavity **29**. This pressure differential **64** generates an inward compressive force **68** that is exerted upon the outer wrapper **14**, the inner liner **16** and the trim breaker **18** in the direction of the insulating cavity **29**. As this inward compressive force **68** is applied on the outer wrapper **14** and inner liner **16**, certain bending, flexion, movement or other deflection of the inner liner **16** and outer wrapper **14** may occur. It is contemplated that the thermal encapsulation material **30** in the sealing state **36** can be at least partially elastic to allow for movement of the liner edge **26** and wrapper edge **22** within the liner and wrapper channels **24**, **20**, respectively, without losing, damaging or otherwise degrading the hermetic seal **48** between the inner liner **16** and the trim breaker **18** and the outer wrapper **14** and the trim breaker **18**. It is contemplated that use of the thermal encapsulation material **30** is particularly useful where the thermal trim breaker **18** is made of a different material than the inner liner **16** and outer wrapper **14**. Typically, the thermal trim breaker **18** will be made of plastic or other similar polymer material, and the inner liner **16** and outer wrapper **14** will each be made of a metallic material.

(17) According to the various embodiments, the structural cabinet **10** that forms the vacuum insulated structure provides for heat transfer between the metal inner liner **16** and the metal outer wrapper **14** with the vacuum insulated structure. When the plastic trim breaker **18** separates the inner liner **16** and outer wrapper **14**, these components typically have a lower heat transfer rate than would be found in a direct metal-to-metal connection between a metallic inner liner **16** and a metallic outer wrapper **14**. The trim breaker **18** is installed at a front face **80** of the structural cabinet **10** and is used as a cap to keep the core materials, such as various insulating materials, inside the insulating cavity **29** between the inner liner **16** and the outer wrapper **14**. The thermal encapsulation material **30** used to attach the inner liner **16** to the trim breaker **18** and the outer wrapper **14** to the trim breaker **18** provides a sturdy connection mechanism for maintaining a sealed engagement between these dissimilar materials and allowing for the generation of an at least partial vacuum **60** within the insulating cavity **29**.

(18) Referring again to FIGS. **1-7**, the use of dissimilar metals in the structural cabinet **10** between

the metallic inner liner **16**, the metallic outer wrapper **14** and the plastic thermal trim breaker **18** results in the metallic portions of the structural cabinet **10** having a different coefficient of thermal expansion than that of the plastic components of the structural cabinet **10**. The elastic nature of the thermal encapsulation material **30** in the sealing state **36** allows for these minimal expansion movements of differing degree between the typically cooler metallic inner liner **16**, the typically warmer outer wrapper **14** and the plastic trim breaker **18** extending therebetween. It is contemplated that the thermal encapsulation material **30** can operate to provide a sturdy hermetic seal **48** between the inner liner **16** and the trim breaker **18** and the outer wrapper **14** and the trim breaker **18** under different temperature environments typically found within and around refrigerating appliances **12**.

(19) According to the various embodiments, it is contemplated that the thermal encapsulation material **30** can be any one of various materials that can include, but are not limited to, thermosetting polymers, thermoplastics, elastomers, combinations thereof, and other similar materials. More specifically, the thermal encapsulation material **30** can include any one or more of various epoxies, silicones, polyurethanes, acrylics, polyimides, silicone polyimides, parylenes, polycycloolefins, silicon-carbons, benzocyclobutenes, liquid crystal polymers, combinations thereof, and other similar encapsulating materials. It is contemplated that the thermal encapsulation material **30** can include first and second components **90**, **92**, and can also include additional distinct components **38** that can be combined to form the application and sealing states **34**, **36** of the thermal encapsulation material **30**.

(20) Referring now to FIGS. **8-10**, it is contemplated that the liner edge **26** can include a plurality of liner protrusions **100** that engage at least one sidewall **102** of the liner channel **24**. Engagement of the plurality of liner protrusions **100** with the liner channel **24** serves to center the liner edge **26** within the liner channel **24**. In this manner, the centering of the liner edge **26** within the liner channel **24** serves to allow the thermal encapsulation material **30** to extend around and engage both of opposing first and second sides **110**, **112** of the liner edge **26**. Similarly, the wrapper edge **22** can include a plurality of wrapper protrusions **104** that engage at least one sidewall **102** of the wrapper channel **20**. As with the liner protrusions **100**, engagement of a plurality of wrapper protrusions **104** with the wrapper channel **20** serves to center the wrapper edge **22** within the wrapper channel **20**. As with the liner protrusions **100** being disposed within the liner channel **24**, the use of the wrapper protrusions **104** serves to center the wrapper edge **22** within the wrapper channel **20** to allow the thermal encapsulation material **30** to be disposed around both of the opposing first and second sides **110**, **112** of the wrapper edge **22**. Through the use of the liner and wrapper protrusions **100**, **104**, the thermal encapsulation material **30** can be allowed to flow or otherwise extend around both sides of the liner edge **26** and the wrapper edge **22**. Accordingly, the thermal encapsulation material **30** surrounds the liner and wrapper edges **22** to fully encapsulate the liner edge **26** and wrapper edge **22** within the liner and wrapper channels **24**, **20**, respectively.

(21) According to the various embodiments, it is contemplated that placement of the liner and wrapper edges **26**, **22** within the liner and wrapper channels **24**, **20**, respectively, can result in the liner and wrapper edges **26**, **22** being free of direct contact with the thermal trim breaker **18**. In such an embodiment, the thermal encapsulation material **30** can completely surround and separate the liner and wrapper edges **26**, **22** from the sidewalls **102** of the liner and wrapper channels **24**, **20**. Accordingly, the thermal encapsulation material **30** provides an additional thermal barrier that slows the degree of thermal transfer between the metallic outer wrapper **14** and the trim breaker **18** and the metallic inner liner **16** and the trim breaker **18**.

(22) According to aspects of the device that include the liner and wrapper protrusions **100**, **104**, as exemplified in FIGS. **8-10**, it is contemplated that the liner and wrapper protrusions **100**, **104** can be substantially hemispheric members that extend from opposing first and second sides **110**, **112**, in an alternating fashion, of each of the inner liner **16** and outer wrapper **14**. The hemisphere configuration of the liner and wrapper protrusions **100**, **104** allows for a minimal contact area

between the liner edge **26** and the liner channel **24** and between the wrapper edge **22** and the wrapper channel **20**. This minimal degree of contact minimizes the amount of thermal transfer that may occur through the direct engagement between the inner liner **16** and the trim breaker **18** and the outer wrapper **14** and the trim breaker **18**.

(23) Referring again to FIGS. **1-7**, after the inner liner **16** and outer wrapper **14** have been disposed within the liner channel **24** and wrapper channel **20** of the trim breaker **18**, respectively, the thermal encapsulation material **30** cures to form the sealing state **36** of the thermal encapsulation material **30**. At this point, gas **62** can be expressed, expelled, or otherwise removed from the insulating cavity **29** of the structural cabinet **10** to define an at least partial vacuum **60** within the structural cabinet **10**. It is contemplated that the insulation material **28** can be disposed between the inner liner **16** and outer wrapper **14** either before or after the trim breaker **18** is attached to the inner liner **16** and the outer wrapper **14**.

(24) Having described various aspects of the structural cabinet **10** using aspects of the thermal encapsulation material **30**, a method **400** is disclosed for forming an insulative structural cabinet **10** for an appliance **12** using aspects of the thermal encapsulation material **30**. According to the method **400**, distinct components **38** of a multi-part thermal encapsulation material **30** are disposed into respective dispensing chambers **40** (step **402**). The separation of the distinct components **38** of the thermal encapsulation material **30** defines a pre-mix state **32** of the thermal encapsulation material **30**. It is contemplated that the distinct components **38** of the thermal encapsulation material **30**, by themselves, can be substantially inert and typically do not serve as a proper adhesive or encapsulation material **30** for the structural cabinet **10**. The distinct components **38** are then delivered from the respective dispensing chambers **40** to the mixing chamber **46** of the material delivery mechanism **42** (step **404**). The distinct components **38** having been disposed in the mixing chamber **46** are then mixed by an impeller **44** within the mixing chamber **46** to define an uncured application state **34** of the thermal encapsulation material **30** (step **406**).

(25) It is contemplated that the application state **34** of the thermal encapsulation material **30** is a substantially fluid state that allows for pouring or flowing of the thermal encapsulation material **30** into the liner channel **24** and wrapper channel **20** of the thermal breaker. It is also contemplated that the application state **34** of the thermal encapsulation material **30** can be a more viscous material that may be injected or otherwise compressed or molded into the liner channel **24** and wrapper channel **20** of the trim breaker **18**.

(26) According to the method **400**, after the components of the thermal encapsulation material **30** have been mixed to define the application state **34**, the thermal encapsulation material **30** in the application state **34** is delivered to the liner channel **24** and the wrapper channel **20** of the trim breaker **18** (step **408**). The wrapper edge **22** of the outer wrapper **14** is then disposed into the wrapper channel **20** so that the thermal encapsulation material **30** in the application state **34** surrounds both sides of the wrapper edge **22** within the wrapper channel **20** (step **410**). Similarly, the liner edge **26** of the inner liner **16** is disposed into the liner channel **24** so that the thermal encapsulation material **30** can surround both sides of the liner edge **26** within the liner channel **24** (step **412**). It is contemplated that steps **410** and **412** can be performed simultaneously or can be switched in order such that the inner liner **16** is first placed within the liner channel **24** and then, subsequently, the outer wrapper **14** is placed within the wrapper channel **20**. After the inner liner **16** and outer wrapper **14** are placed within the thermal encapsulation material **30** in the application state **34**, the thermal encapsulation material **30** is then cured within the wrapper and liner channels **20**, **24** to define a sealing state **36** of the thermal encapsulation material **30** (step **414**). As discussed above, the thermal encapsulation material **30** in the sealing state **36** defines a hermetic seal **48** between the inner liner **16** and the trim breaker **18** and between the outer wrapper **14** and the trim breaker **18**.

(27) Referring again to FIGS. **1-7**, after formation of the structural cabinet **10**, the insulation material **28** can be disposed within an insulating cavity **29** defined between the inner liner **16** and

the outer wrapper **14**. Gas **62** can then be expressed from the insulating cavity **29** and from the insulation material **28** defined within the insulating cavity **29**. This expression of gas **62** serves to define the at least partial vacuum **60** within the insulating cavity **29**. It is contemplated that the thermal encapsulation material **30** in the sealing state **36** permits at least partial movement of the inner liner **16** and outer wrapper **14** during and after the expression of gas **62** from the insulating cavity **29**. The encapsulation material **30** allows for this partial movement while also maintaining a hermetic seal **48** between the inner liner **16** and the trim breaker **18** and between the outer wrapper **14** and the trim breaker **18**.

(28) Referring again to FIGS. **1-7**, it is contemplated that the thermal encapsulation material **30** can include first and second components **90**, **92** that are disposed within respective first and second dispensing chambers **120**, **122** of the material delivery mechanism **42**, when the thermal encapsulation material **30** is in a pre-mix state **32**.

(29) Referring again to FIGS. **9-11**, it is contemplated that the step **412** of disposing the inner liner **16** within the liner channel **24** can include centering the liner edge **26** within the liner channel **24** using the liner protrusions **100** that engage at least one sidewall **102** within the liner channel **24**. As discussed above, engagement plurality of liner protrusions **100** with the liner channel **24** serves to center the liner edge **26** within the liner channel **24** such that the thermal encapsulation material **30** engages both sides of the liner edge **26**. Similarly, the step **410** of disposing the wrapper edge **22** within the wrapper channel **20** includes centering the wrapper edge **22** within the wrapper channel **20** using the wrapper protrusions **104**. As with the liner protrusions **100**, the wrapper protrusions **104** serve to center the wrapper edge **22** within the wrapper channels **20** such that the thermal encapsulation material **30** engages both sides of the wrapper edge **22**.

(30) According to the various embodiments, it is contemplated that the liner and wrapper protrusions **100**, **104** can be hemispheric projections, dimples, detents, indents, combinations thereof, and other similar protrusions. Typically, the protrusions will be pressed or punched formations within the surface of the inner liner **16** and outer wrapper **14** such that one side of the protrusion projects outward from one surface of the inner liner/outer wrapper **16**, **14** and the opposing side of the outward protrusion **126** defines an inward indent **128** within the opposing first and second sides **110**, **112** of each of the inner liner/outer wrapper **16**, **14**. These geometries of the liner and wrapper protrusions **100**, **104** can also serve to increase the retaining force of the thermal encapsulation material **30** with the inner liner **16** and outer wrapper **14**. Stated another way, the liner protrusion **100** defines the outward protrusion **126** on a first liner surface **130** and the inward indent **128** on an opposing second liner surface **132**. Similarly, the wrapper protrusions **104** can define the outward protrusion **126** on a first wrapper surface **134** and the inward indent **128** on a second wrapper surface **136**. In this manner, the wrapper and liner protrusions **104**, **100** define alternating outward protrusions **126** and inward indents **128** along the opposing surfaces of the inner liner **16** and outer wrapper **14**.

(31) Referring again to FIGS. **1-10** and **12**, a method **500** is disclosed for forming an insulating structural cabinet **10** for an appliance **12** using an aspect of the thermal encapsulation material **30**. According to the method **500**, first and second components **90**, **92** of a thermal encapsulation material **30** are disposed into respective first and second dispensing chambers **120**, **122** to define a pre-mix state **32** of the thermal encapsulation material **30** (step **502**). The first and second components **90**, **92** are then delivered from the respective first and second dispensing chambers **120**, **122** to a mixing chamber **46** (step **504**). The thermal encapsulation material **30** is activated by combining the first and second components **90**, **92** of the thermal encapsulation material **30** within the mixing chamber **46** to generate a chemical reaction that defines an application state **34** of the thermal encapsulation material **30** (step **506**).

(32) As discussed above, the application state **34** of the thermal encapsulation material **30** can be a fluid and substantially flowable material or can be a more viscous and injectable material that can be disposed within the liner and wrapper channels **24**, **20** of the trim breaker **18**.

(33) According to the method **500**, the thermal encapsulation material **30** is then disposed, while in the application state **34**, into the liner channel **24** and the wrapper channel **20** of the trim breaker **18** (step **508**). A portion of the inner liner **16** is then positioned, and typically centered, within the thermal encapsulation material **30** in the liner channel **24** (step **510**). A portion of the outer wrapper **14** is then positioned and typically centered, within the thermal encapsulation material **30** within the wrapper channel **20** (step **512**). The thermal encapsulation material **30** is then cured around the portions of the inner liner **16** and outer wrapper **14** that are disposed within the liner and wrapper channels **24**, **20**, respectively (step **514**). The cured thermal encapsulation material **30** defines the sealing state **36** of the thermal encapsulation material **30** that is characterized by a hermetic seal **48** between the inner liner **16** of the trim breaker **18** and between the outer wrapper **14** of the trim breaker **18**.

(34) As discussed above, each of the inner liner **16** and outer wrapper **14** can include positioning features, typically in the form of the liner and wrapper protrusions **100**, **104**, that at least partially engage the liner channel **24** and the wrapper channel **20**, respectively. It is contemplated that the positioning features are adapted to centrally align the inner liner **16** within the liner channel **24** and to also centrally align the outer wrapper **14** within the wrapper channel **20**. According to various embodiments, it is contemplated that the positioning features can define minimal contact between the liner and wrapper edges **26**, **22** and the liner and wrapper channels **24**, **20**, respectively. It is also contemplated that the positioning features can be configured to space the liner and wrapper edges **26**, **22** away from the sidewalls **102** of the liner and wrapper channels **24**, **20**, respectively. In such an embodiment, the liner and wrapper edges **26**, **22** are free of direct engagement with the liner and wrapper channels **24**, **20** and are fully separated by the thermal encapsulation material **30**.

(35) According to the various embodiments, it is contemplated that the thermal encapsulation material **30** can be used in the formation of structural cabinets **10** for various appliances **12**. These appliances **12** can include, but are not limited to, refrigerators, freezers, coolers, ovens, dishwashers, laundry appliances, water heaters, and other similar appliances **12** and fixtures within household and commercial settings.

(36) It will be understood by one having ordinary skill in the art that construction of the described device and other components is not limited to any specific material. Other exemplary embodiments of the device disclosed herein may be formed from a wide variety of materials, unless described otherwise herein.

(37) For purposes of this disclosure, the term “coupled” (in all of its forms, couple, coupling, coupled, etc.) generally means the joining of two components (electrical or mechanical) directly or indirectly to one another. Such joining may be stationary in nature or movable in nature. Such joining may be achieved with the two components (electrical or mechanical) and any additional intermediate members being integrally formed as a single unitary body with one another or with the two components. Such joining may be permanent in nature or may be removable or releasable in nature unless otherwise stated.

(38) It is also important to note that the construction and arrangement of the elements of the device as shown in the exemplary embodiments is illustrative only. Although only a few embodiments of the present innovations have been described in detail in this disclosure, those skilled in the art who review this disclosure will readily appreciate that many modifications are possible (e.g., variations in sizes, dimensions, structures, shapes and proportions of the various elements, values of parameters, mounting arrangements, use of materials, colors, orientations, etc.) without materially departing from the novel teachings and advantages of the subject matter recited. For example, elements shown as integrally formed may be constructed of multiple parts or elements shown as multiple parts may be integrally formed, the operation of the interfaces may be reversed or otherwise varied, the length or width of the structures and/or members or connector or other elements of the system may be varied, the nature or number of adjustment positions provided between the elements may be varied. It should be noted that the elements and/or assemblies of the

system may be constructed from any of a wide variety of materials that provide sufficient strength or durability, in any of a wide variety of colors, textures, and combinations. Accordingly, all such modifications are intended to be included within the scope of the present innovations. Other substitutions, modifications, changes, and omissions may be made in the design, operating conditions, and arrangement of the desired and other exemplary embodiments without departing from the spirit of the present innovations.

(39) It will be understood that any described processes or steps within described processes may be combined with other disclosed processes or steps to form structures within the scope of the present device. The exemplary structures and processes disclosed herein are for illustrative purposes and are not to be construed as limiting.

(40) It is also to be understood that variations and modifications can be made on the aforementioned structures and methods without departing from the concepts of the present device, and further it is to be understood that such concepts are intended to be covered by the following claims unless these claims by their language expressly state otherwise.

(41) The above description is considered that of the illustrated embodiments only. Modifications of the device will occur to those skilled in the art and to those who make or use the device. Therefore, it is understood that the embodiments shown in the drawings and described above is merely for illustrative purposes and not intended to limit the scope of the device, which is defined by the following claims as interpreted according to the principles of patent law, including the Doctrine of Equivalents.

Claims

1. A trim breaker assembly for an appliance, the trim breaker assembly comprising: a trim breaker having a wrapper channel configured to receive a wrapper edge of an outer wrapper and a liner channel configured to receive a liner edge of an inner liner; a multi-component thermal encapsulation material defining a pre-mix state, an application state and a sealing state, the pre-mix state defined by distinct components of the multi-component thermal encapsulation material being separated from one another, the application state defined by the distinct components combined together into an uncured state of the multi-component thermal encapsulation material, and the sealing state defined by the multi-component thermal encapsulation material disposed within the wrapper and liner channels and surrounding the wrapper and liner edges, respectively, in the sealing state that is configured to define a seal between the trim breaker and the wrapper edge and the liner edge; and a centering assembly that is configured to centrally position the wrapper edge within wrapper channel sidewalls of the wrapper channel of the trim breaker and to centrally position the liner edge within liner channel sidewalls of the liner channel of the trim breaker.
2. The trim breaker assembly of claim 1, wherein the trim breaker is made of plastic.
3. The trim breaker assembly of claim 1, wherein the multi-component thermal encapsulation material includes at least one of epoxy, silicone, polyurethane, acrylic, polyimide, silicone polyimide, parylene, polycycloolefin, silicon-carbon, benzocyclobutene, and a liquid crystal polymer.
4. The trim breaker assembly of claim 1, wherein the multi-component thermal encapsulation material is a thermosetting polymer.
5. The trim breaker assembly of claim 1, wherein the multi-component thermal encapsulation material is configured to engage opposing sides of the inner liner and both sides of the outer wrapper.
6. The trim breaker assembly of claim 1, wherein the wrapper channel sidewalls and the liner channel sidewalls at least partially define a sidewall of the trim breaker that separates the liner channel from the wrapper channel.
7. The trim breaker assembly of claim 1, wherein the centering assembly is configured to be

attached to the liner edge of the inner liner.

8. The trim breaker assembly of claim 1, wherein the centering assembly is configured to be attached to the wrapper edge of the outer wrapper.

9. The trim breaker assembly of claim 1, wherein the distinct components of the multi-component thermal encapsulation material includes a first component and a second component.

10. The trim breaker assembly of claim 1, wherein the seal is a hermetic seal.

11. A trim breaker assembly for an appliance, the trim breaker assembly comprising: a trim breaker having wrapper channel sidewalls that define a wrapper channel and liner channel sidewalls that define a liner channel; a multi-component thermal encapsulation material defining a pre-mix state, an application state and a sealing state, the pre-mix state defined by distinct components of the multi-component thermal encapsulation material being separated from one another, the application state defined by the distinct components combined together into an uncured state of the multi-component thermal encapsulation material, and the sealing state defined by the multi-component thermal encapsulation material disposed within the wrapper channel and the liner channel and surrounding the wrapper edge and the liner edge, respectively, in the sealing state that defines a seal between the trim breaker and the wrapper edge and the liner edge; a wrapper edge that is configured to be disposed within the multi-component thermal encapsulation material within the wrapper channel; a liner edge that is configured to be disposed within the multi-component thermal encapsulation material within the liner channel; and a centering assembly that is configured to centrally position the wrapper edge between the wrapper channel sidewalls of the wrapper channel and to centrally position the liner edge between the liner channel sidewalls of the liner channel.

12. The trim breaker assembly of claim 11, wherein the multi-component thermal encapsulation material is a thermosetting polymer.

13. The trim breaker assembly of claim 11, wherein the multi-component thermal encapsulation material is configured to engage opposing sides of the liner edge and both sides of the wrapper edge.

14. The trim breaker assembly of claim 11, wherein the wrapper channel sidewalls and the liner channel sidewalls at least partially define a sidewall of the trim breaker that separates the liner channel from the wrapper channel.

15. The trim breaker assembly of claim 11, wherein the centering assembly is configured to be attached to the liner edge of an inner liner.

16. The trim breaker assembly of claim 11, wherein the centering assembly is configured to be attached to the wrapper edge of an outer wrapper.

17. The trim breaker assembly of claim 11, wherein the distinct components of the multi-component thermal encapsulation material include a first component and a second component.

18. The trim breaker assembly of claim 11, wherein the seal is a hermetic seal.

19. A trim breaker assembly for an appliance, the trim breaker assembly comprising: a trim breaker having wrapper channel sidewalls that define a wrapper channel configured to receive a wrapper edge of an outer wrapper and liner channel sidewalls that define a liner channel configured to receive a liner edge of an inner liner; and a multi-component thermal encapsulation material defining a pre-mix state, an application state and a sealing state, the pre-mix state defined by distinct components of the multi-component thermal encapsulation material being separated from one another, the application state defined by the distinct components combined together into an uncured state of the multi-component thermal encapsulation material, and the sealing state defined by the multi-component thermal encapsulation material disposed within the wrapper and liner channels and surrounding the wrapper and liner edges, respectively, in the sealing state that is configured to define a seal between the trim breaker and the wrapper edge and the liner edge, wherein at least one of the liner edge and the wrapper edge includes a centering assembly that centrally positions the wrapper edge within the wrapper channel sidewalls of the wrapper channel of the trim breaker and centrally positions the liner edge within the liner channel sidewalls of the

liner channel of the trim breaker.

20. The trim breaker assembly of claim 19, wherein the centering assembly includes a plurality of liner protrusions that extend outward from opposing surfaces of the liner edge, wherein a portion of the plurality of liner protrusions extend outward from a first liner surface and extend inward into a second liner surface, and wherein the centering assembly includes a plurality of wrapper protrusions that extend outward from opposing surfaces of the wrapper edge, wherein a portion of the plurality of wrapper protrusions extend outward from a first wrapper surface and extend inward into a second wrapper surface.
